Cognition and Perception

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Neuro Rehabilitation

Guest Lecturer

Learning outcomes

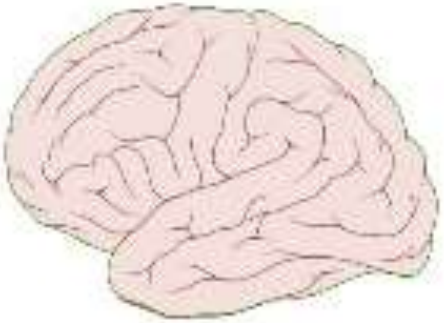
- Explain the role of cognition and perception in functional movement and rehabilitation outcomes.
- Identify common cognitive impairments (e.g., attention, memory, executive function) and perceptual deficits (e.g., neglect, apraxia, agnosia) in neurological populations.
- Select and justify appropriate assessment tools to evaluate cognition and perception in clinical practice.
- Discuss physiotherapy strategies to address cognitive and perceptual problems in rehabilitation.
- Integrate cognitive and perceptual considerations into patient assessment and treatment planning.

Perception is the term used to refer to the processes involved in making sense of the senses; the way in which we interpret the information gathered and processed by the senses.

Cognition is the ability to perceive, represent and organise objects, events and their relationship to one another in an appropriate way. It comprises all the mental processes that allow us to attend to, recognise, learn and remember changing information in the environment.

input → processing → output







perception → cognition → action

(thinking and memory) (motor system)

Goal orientated activity

Find the number

Remember it long enough to dial it

Produce and understand spoken words



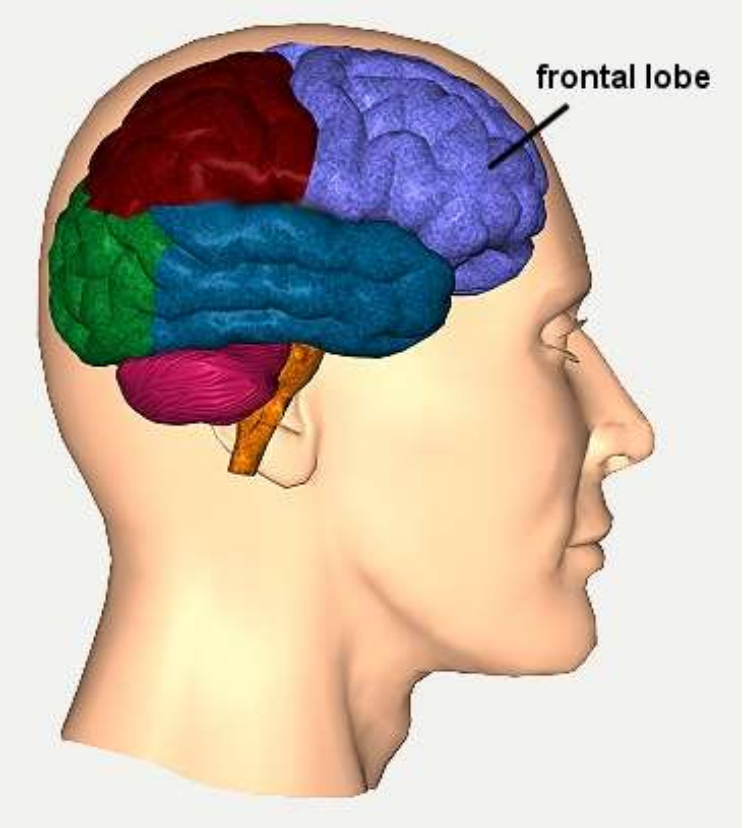
perception

memory

motor planning

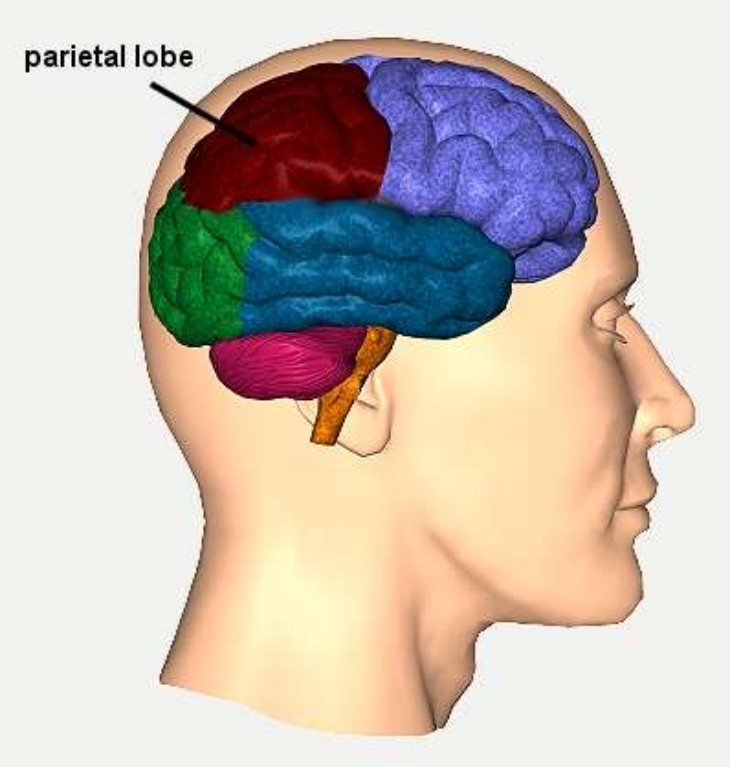
sustained attention

Frontal lobe



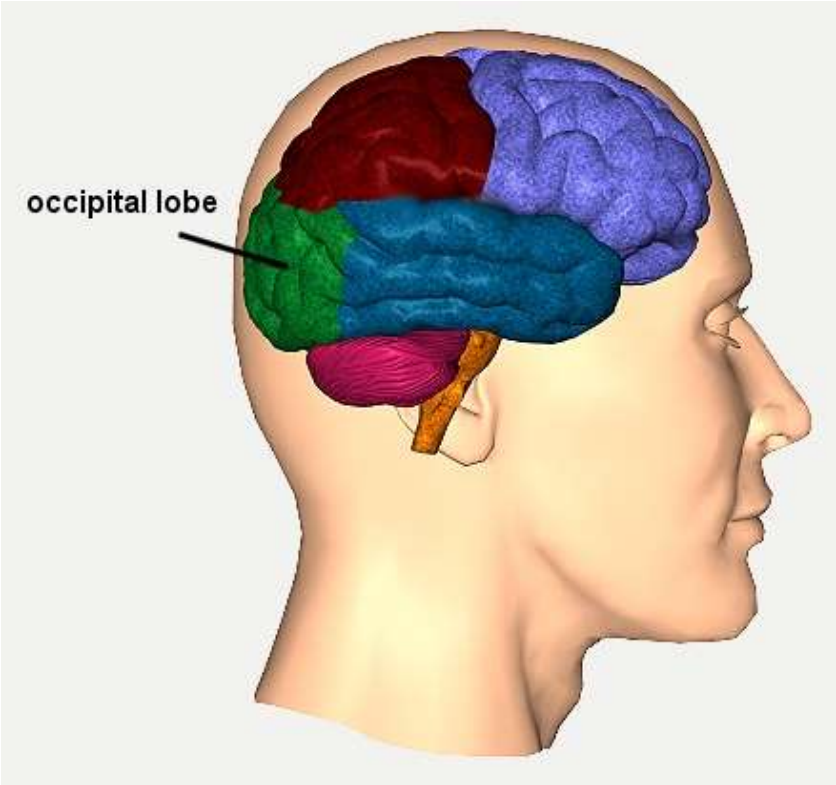
- Attention
- Higher level cognitive function
- Memory

Parietal lobe



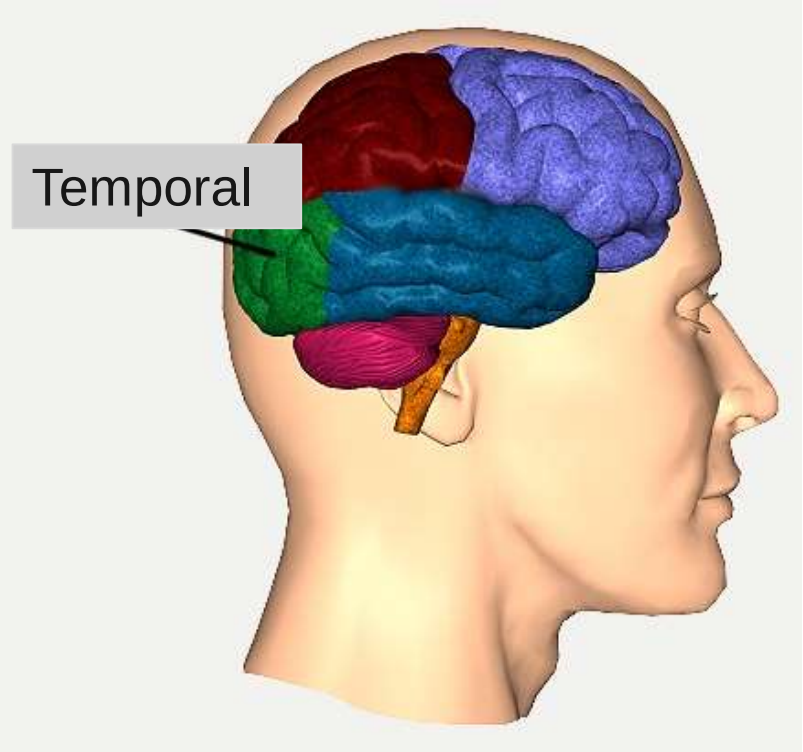
- Interpretation of sensory input
- Contralateral sensation weight, pressure, pain, etc.
- Spatial recognition “where” processing

Occipital lobe



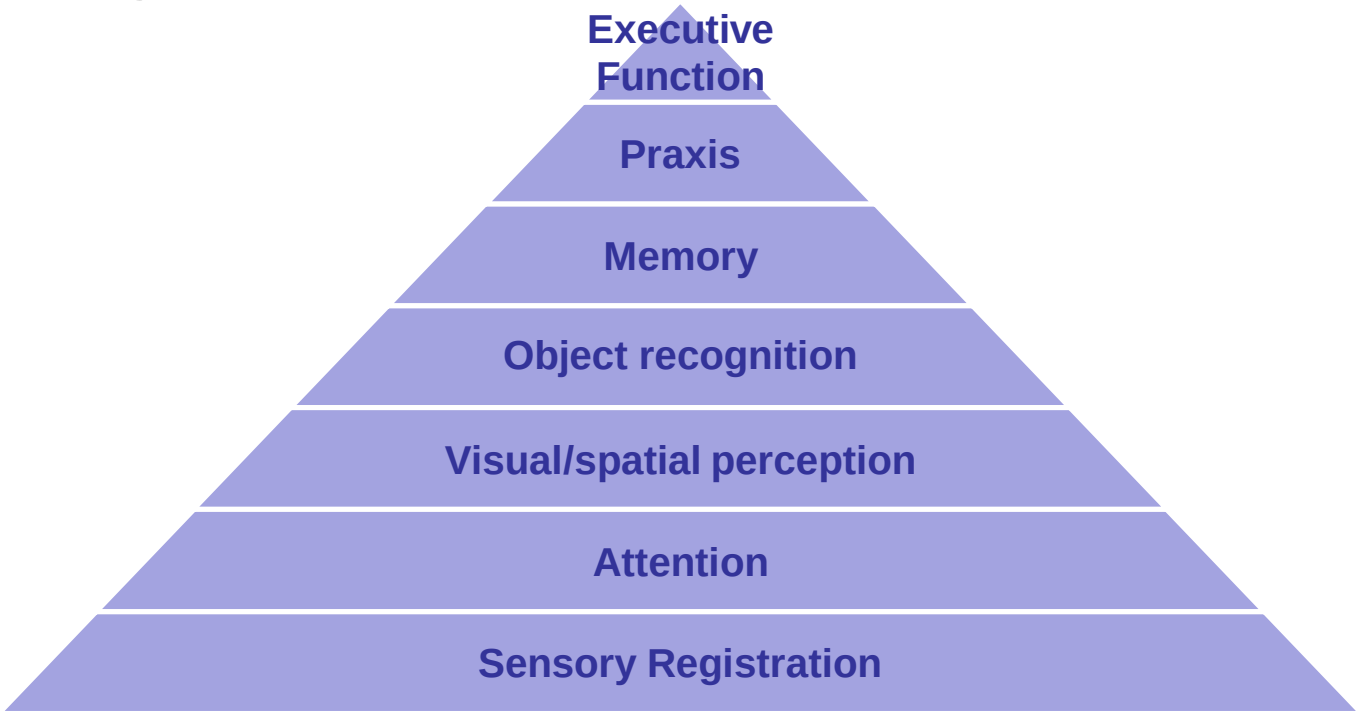
- Primary visual area
- Analyses colour, form, location, motion

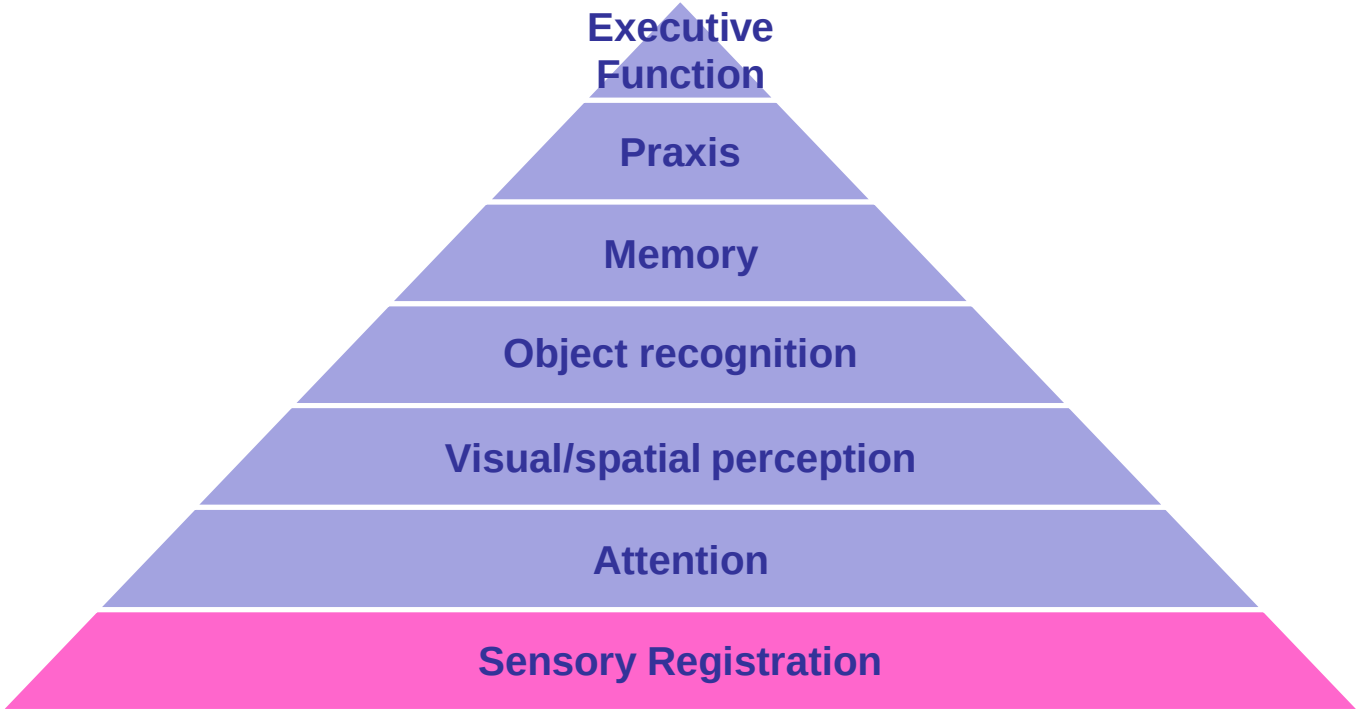
Temporal lobe



- Auditory and olfactory area
- Detailed memories
- Wernicke’s area
- Processes visual information of objects “what”

Hierarchy of cognition




Executive Function
Praxis
Memory
Object recognition
Visual/spatial perception
Attention
Sensory Registration

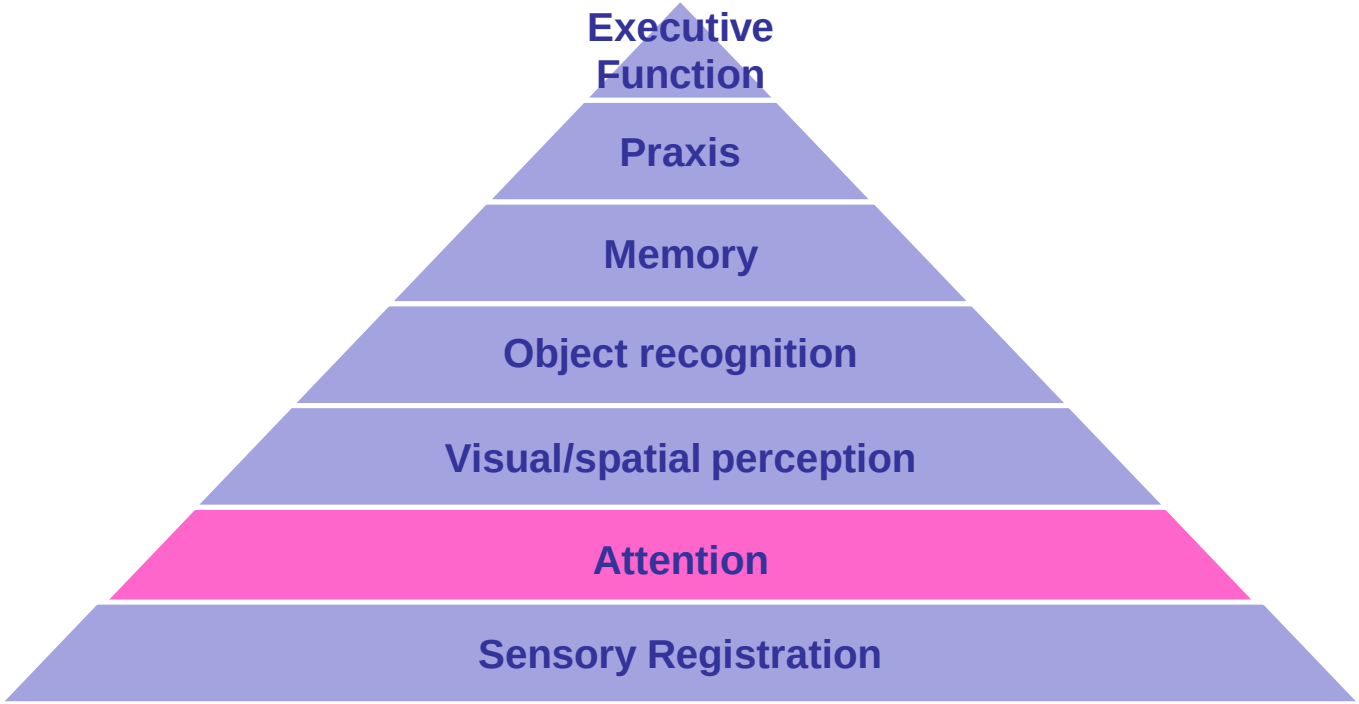
Sensory registration

Information reaches the nervous system from the sense organs: sight, hearing, touch, smell, taste (proprioception)

External physical stimuli are converted to a form that the nervous system can use – coding/registration

Vision is the dominant sense in cognition

- Visual assessment:
  - Visual acuity (normally wear glasses)
  - Tracking and scanning
  - Visual fields: confrontation test


Executive Function
Praxis
Memory
Object recognition
Visual/spatial perception
Attention
Sensory Registration

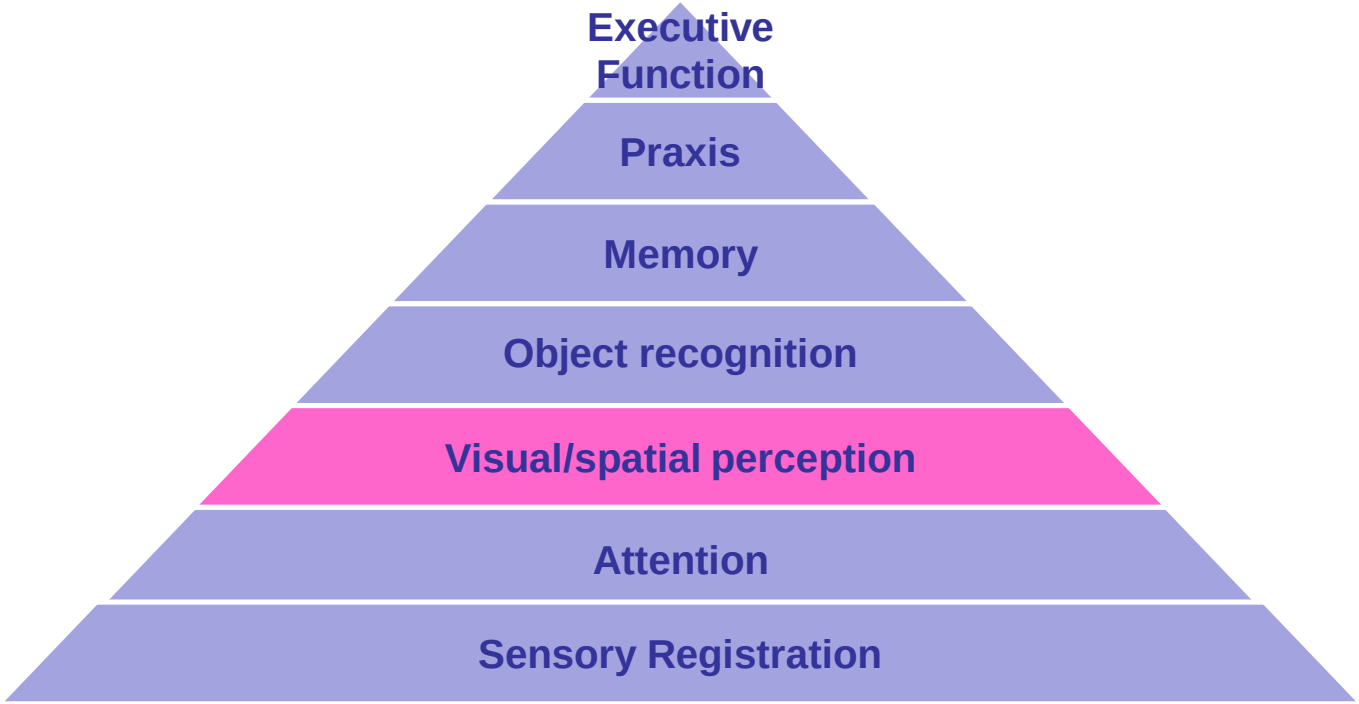
Attention

Attention is a basic part of processing- prerequisites include alertness and arousal

Focuses cognitive resources on selected stimuli from sensory registration.

Attention is affected by internal (fatigue, thirst) and external influences (temperature), individual differences and by task novelty (new tasks increase attentional demands)

Type of Attention	Description	Example in Physiotherapy
Focused	Respond to specific stimuli	Reacting to verbal/tactile cue
Sustained	Maintain attention over time	Staying on task during exercises
Selective	Filter distractions	Focusing in busy rehab gym
Alternating	Switch attention between tasks	Listening then performing
Divided	Do two tasks at once	Walking + talking


Executive Function
Praxis
Memory
Object recognition
Visual/spatial perception
Attention
Sensory Registration

Visual and spatial perception

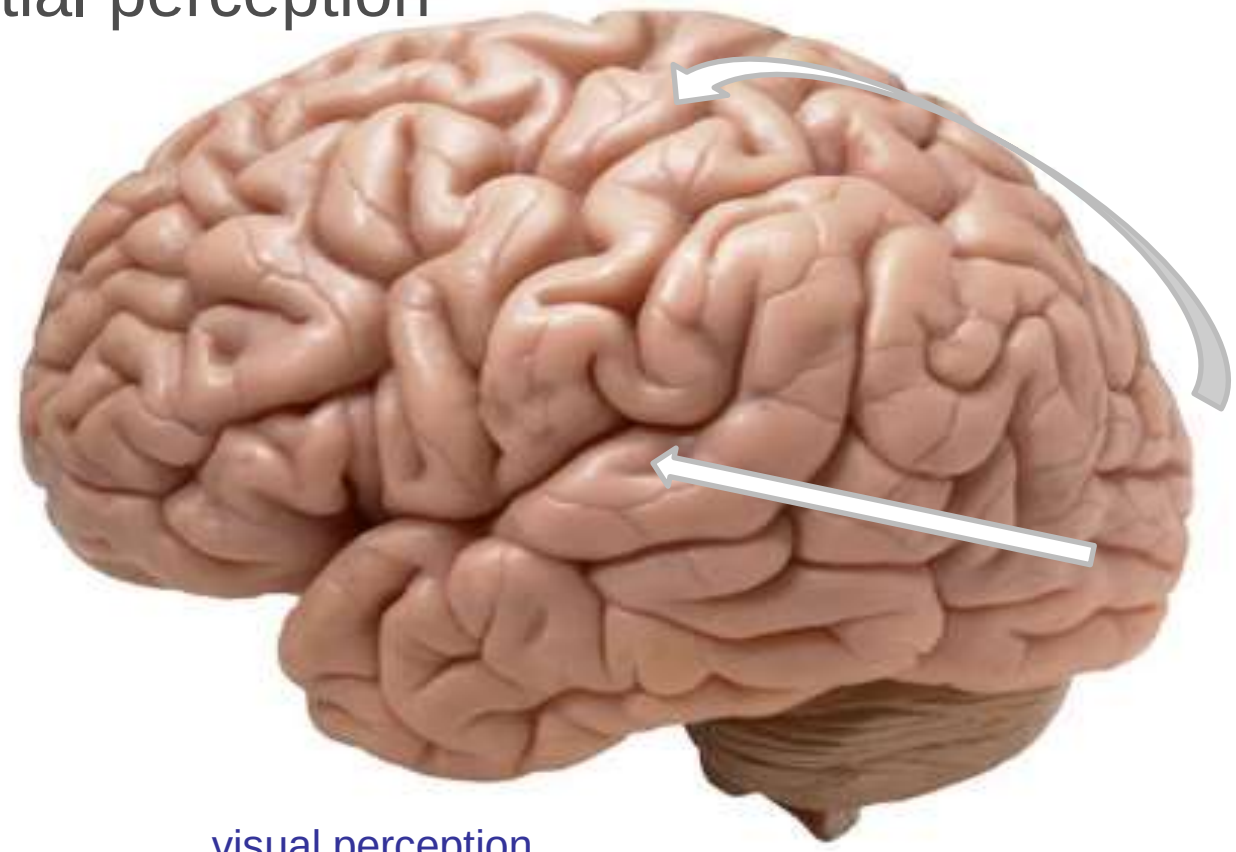
Complex mechanism in which sensory information is given meaning through association with our stored knowledge

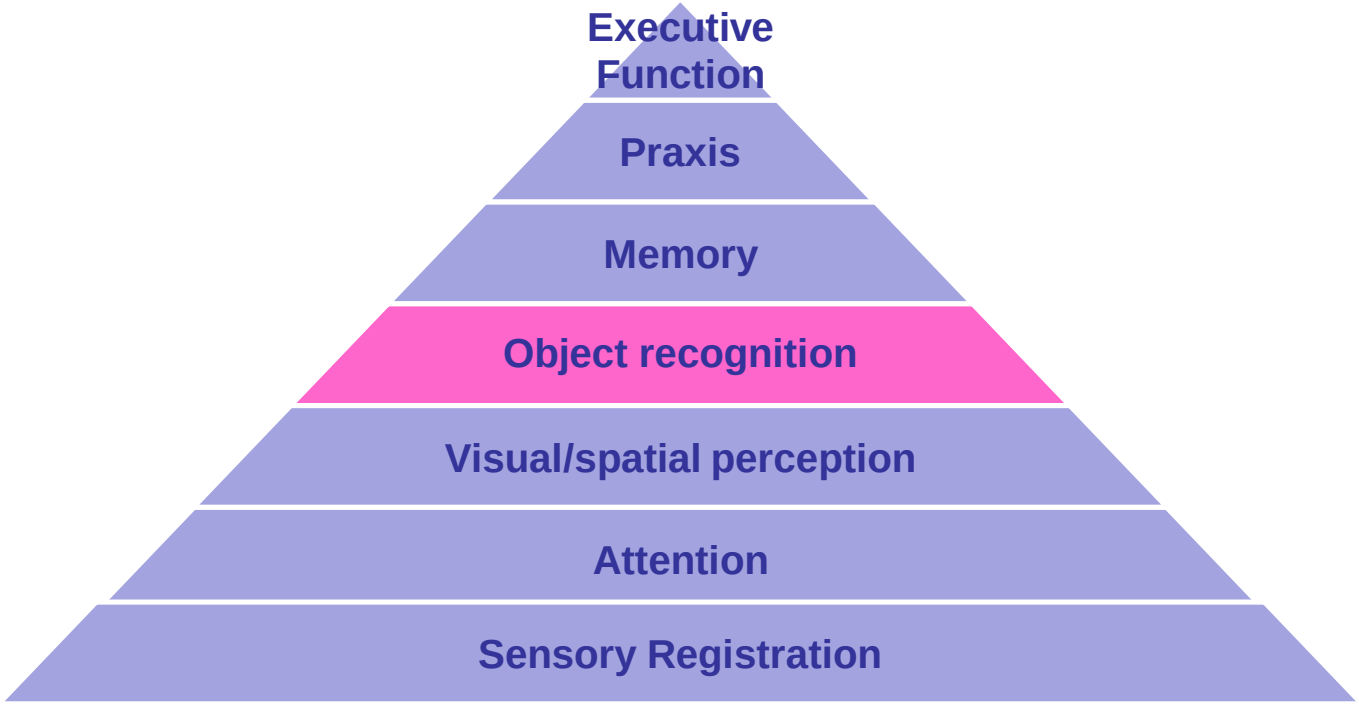
Divided into visual perception and visuospatial perception

- Visuoperceptive skills include colour perception, depth perception and figure ground perception (What things are)

- Visuospatial perception skills includes the ability to appreciate the spatial relationships between objects and landmarks (Where things are)

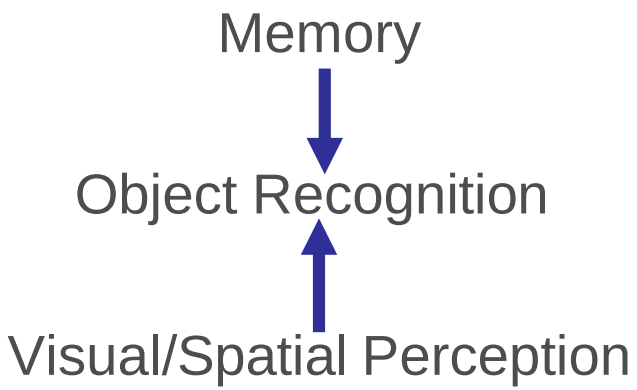
Visual and spatial perception




Executive Function
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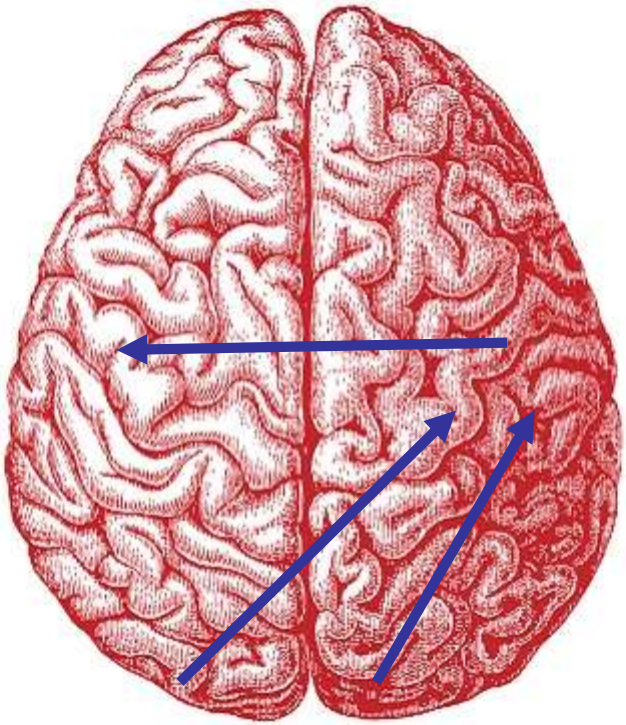
Object recognition

Object recognition integrates basic perception with stored knowledge of objects and their use

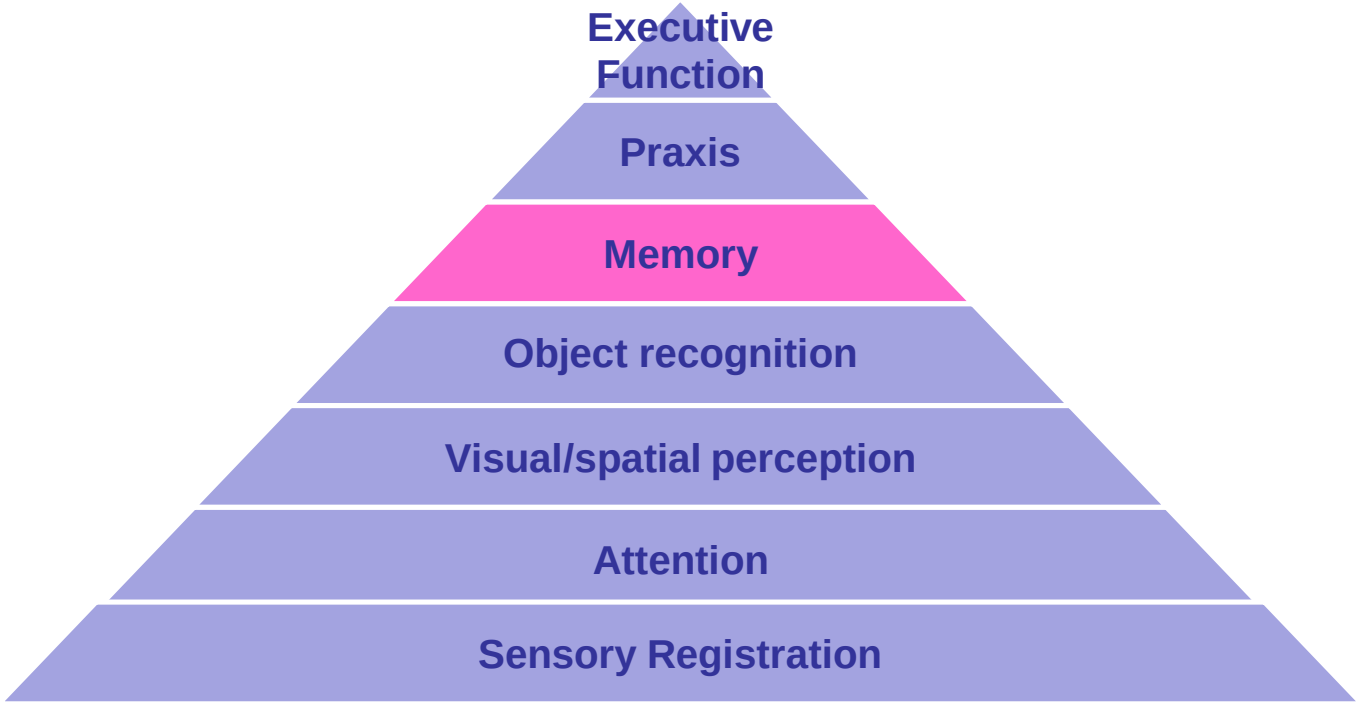


Object recognition

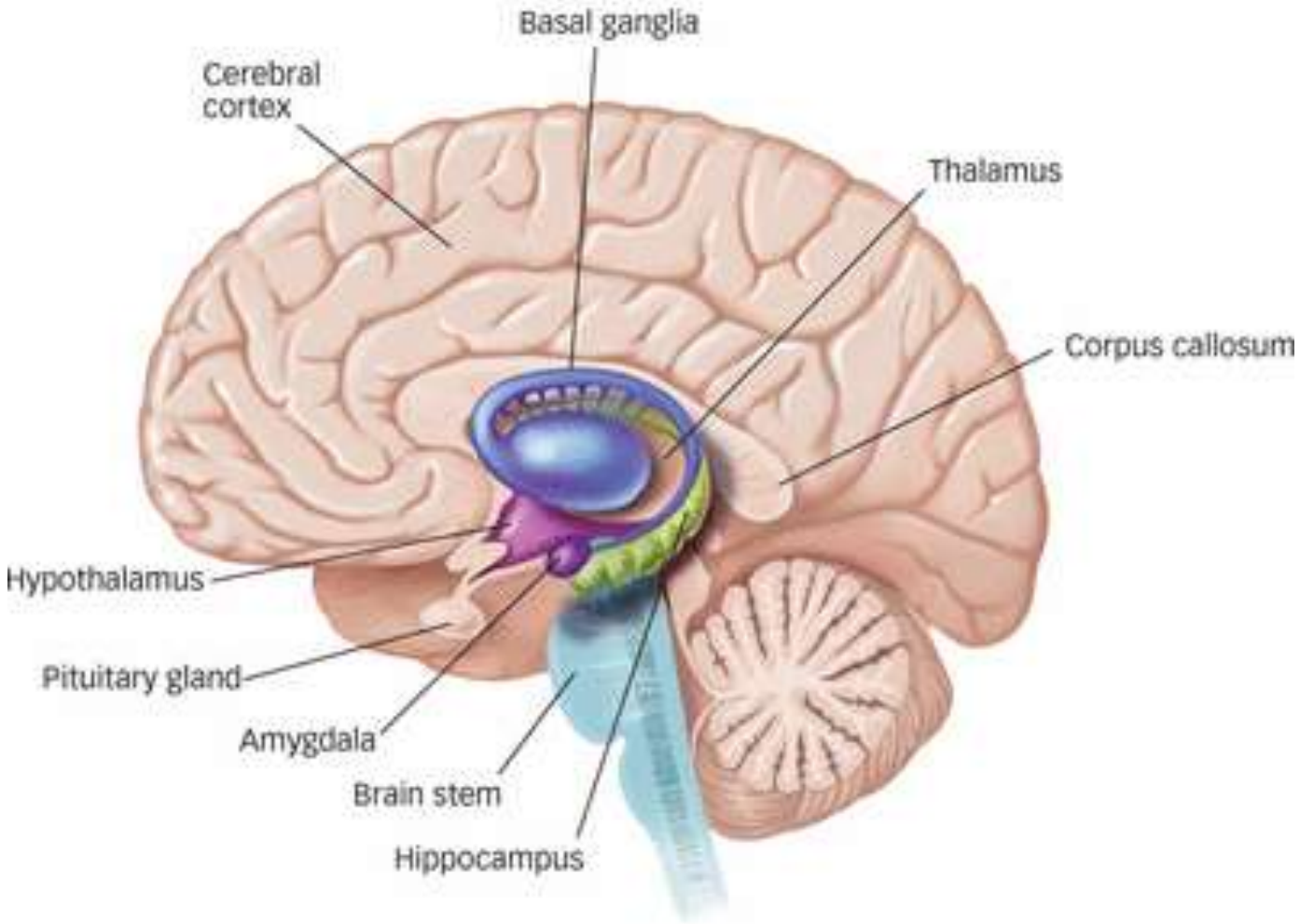
associative recognition
(L temporal)



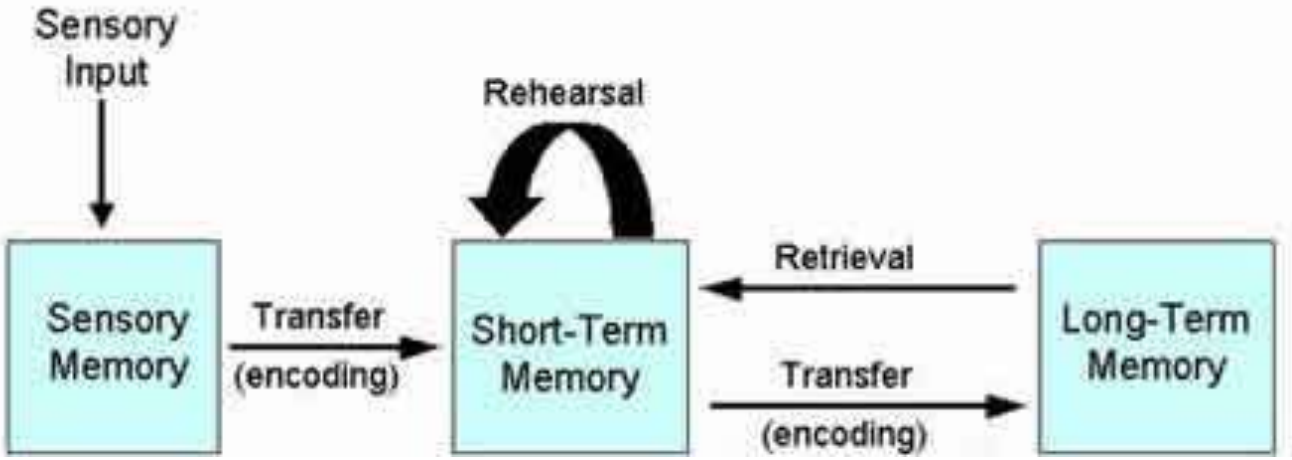
perceptual recognition
(R parietal/temporal)


Executive Function
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Memory


Basal ganglia
Cerebral cortex
Thalamus
Corpus callosum
Hypothalamus
Pituitary gland
Amygdala
Brain stem
Hippocampus

Memory

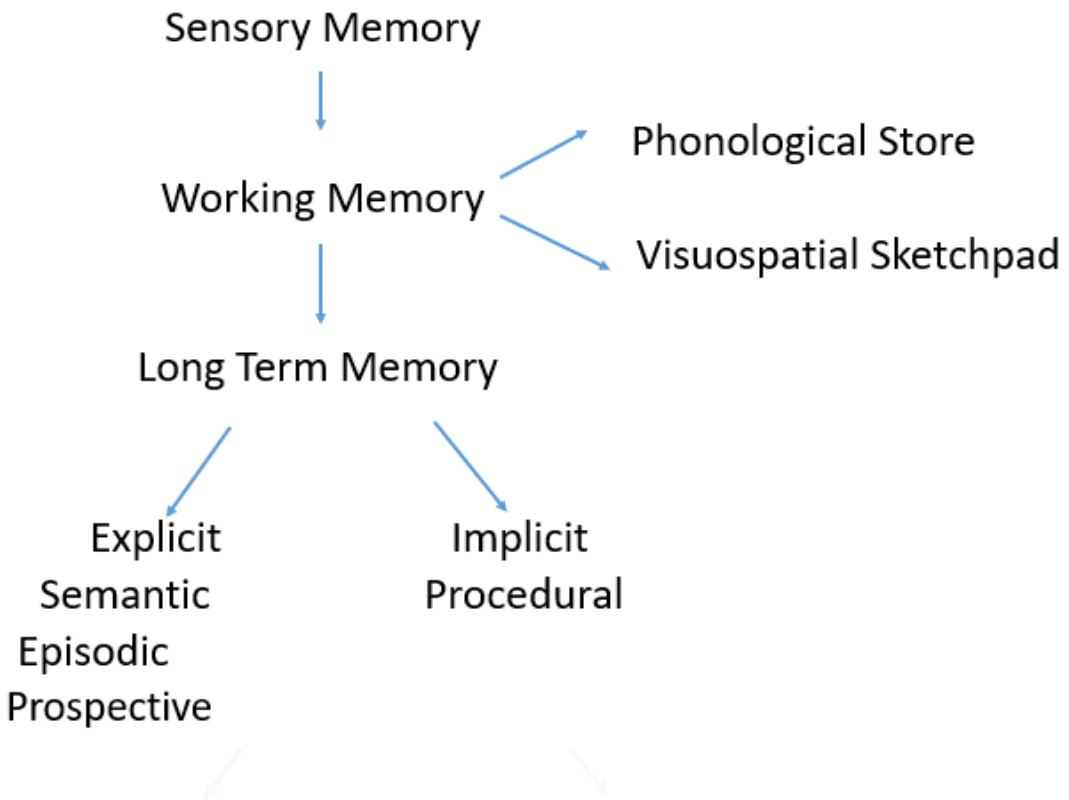


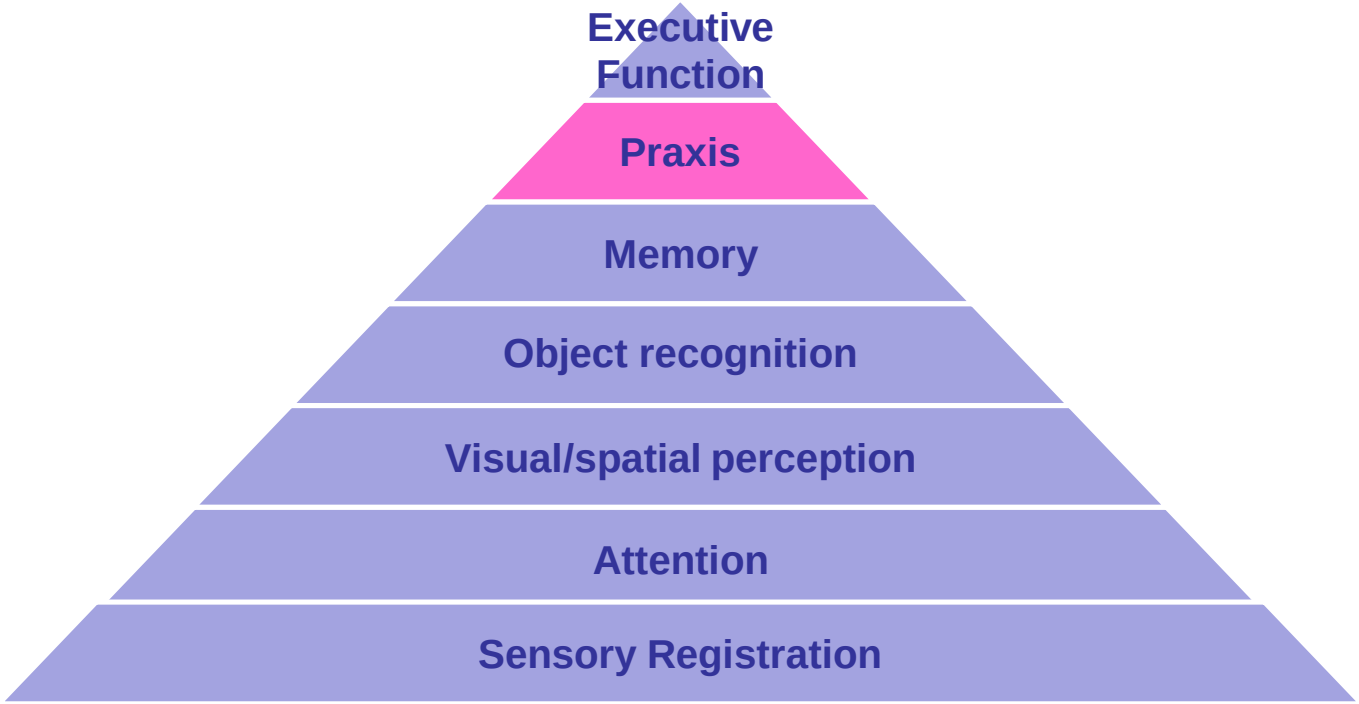
Memory allows us to retain and recall information on all aspects of daily life.

Stores and retrieves information from experience.
Short-term (working memory): holds active task information (e.g., remembering movement sequence).
Long-term: consolidates learning and skills.

3 stages of memory
- encoding (transfer stimulus into a code that the brain accepts)
- storage (retaining information)
- retrieval (recall or recognition: accessing stored information)

Memory


Sensory Memory
Working Memory
Phonological Store
Visuospatial Sketchpad
Long Term Memory
Explicit
Semantic
Episodic
Prospective
Implicit
Procedural


Executive Function
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Praxis

Praxis refers to the ability to use our hands and body in skilled/previously learned tasks in order to function

Praxis involves three component systems (which can be selectively impaired with neurological damage)

Praxis

Sensory/Perceptual system

Provides
Visual/Auditory/
Object information



Conceptual System

Knowledge of
- object use
- action
- sequencing action

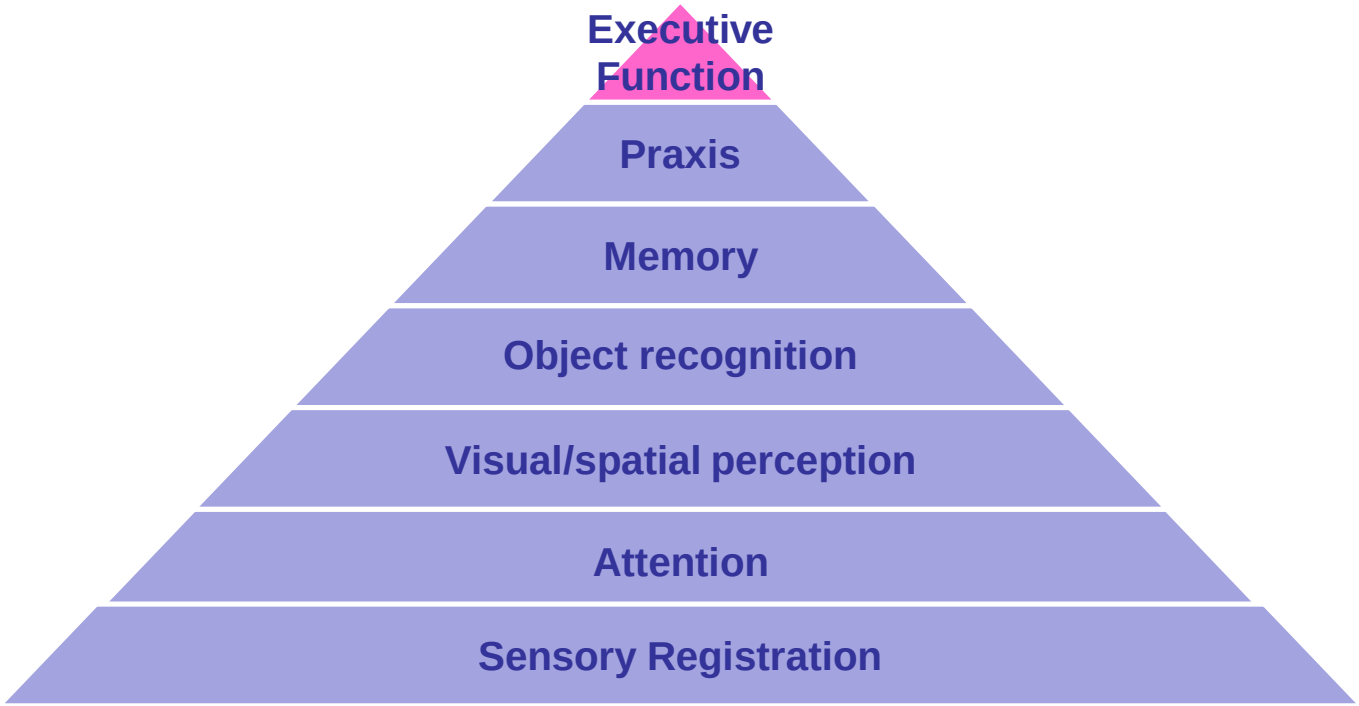


Production System

Involves
- motor planning
- execution of action



Stage	Description	Example
1 Ideation	Concept of the movement or task	Knowing that a toothbrush is used to clean teeth
2 Planning (Ideomotor stage)	Selecting and sequencing appropriate motor actions	Planning the steps — pick up brush, apply toothpaste, move arm to mouth
3 Execution	Carrying out the planned movement	Physically performing the brushing action


Executive Function
Praxis
Memory
Object recognition
Visual/spatial perception
Attention
Sensory Registration

Executive functions

- Comprises the mental functions necessary for formulating goals, planning, how to achieve them and carrying out plans effectively

- Eight items are considered key skills:

- realistic goal setting
- planning
- organisation
- self-initiation
- self-directing
- self-inhibiting
- self-monitoring
- flexible problem solving

Executive function

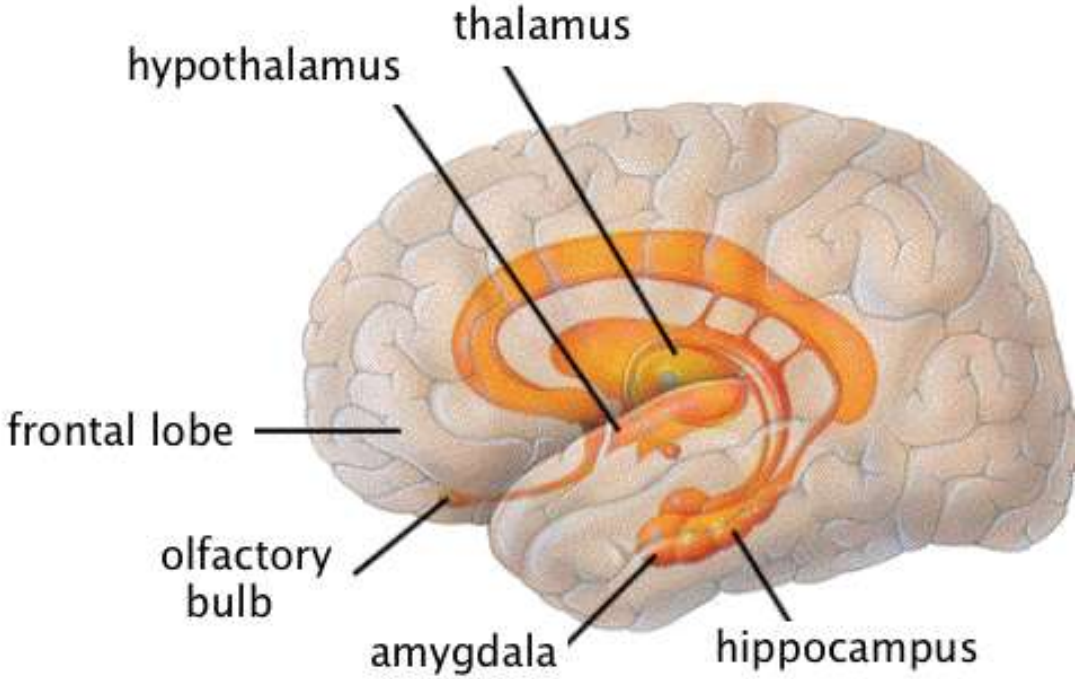
- Executive dysfunction can lead to difficulty formulating goals and maintaining behaviour to complete tasks successfully

- People with executive problems may appear disorganised and lack planning skills, they may not have insight into their difficulties and may have problems interacting appropriately in various social situations.

Limbic system

Executive function

Emotion



Visual and Spatial Perception

- The ability to accurately perceive relationships of objects and oneself in space.

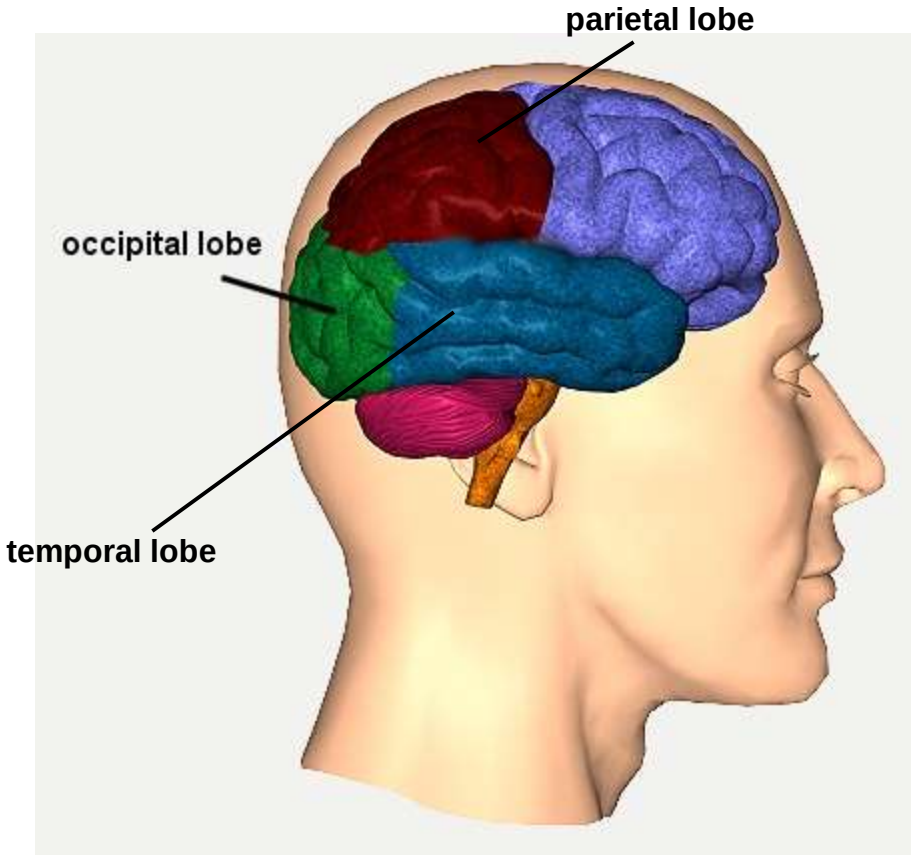
- Visual perception gives meaning to all information entering our eyes.

- It’s NOT a problem with vision, but the interpretation of the visual information

Three areas of visual and spatial perceptual impairments

Body scheme disorder	Knowledge and representation of body parts in relation to each other and within
Visual discrimination	The ability to recognise details in visual images – colour, shape, form, depth, size of objects, people etc.
Object recognition	Ability to interpret visual information to recognise objects, people etc.

Visual and Spatial Perception



Neglect/Inattention

- A body scheme disorder

- An attention deficit but to one side of space/ body part. More common on left side, i.e. with right sided lesion.

- Patient may fail to respond to objects or stimuli on their affected side. May be visual, sensory, or both.

Topographical disorientation (occipital, temporal, parietal lobes)

Inability to find your way in space.

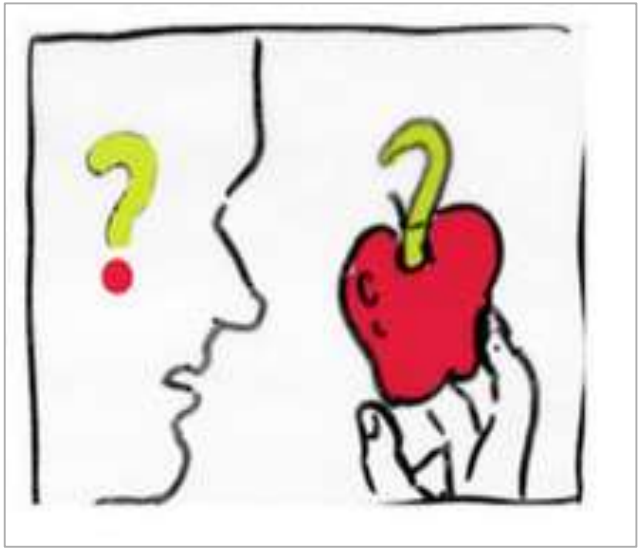
Multiple visual discrimination skills required; distance of objects, colour, shape, form.

Patients may not be able to navigate within familiar environments or on new routes that they have practiced multiple times.



Agnosia (occipital, parietal lobes)

- The inability to identify an object.
- Can be only certain objects e.g faces
- Can be broad e.g. various everyday objects.
- May struggle with functional tasks because ***they do not recognise the tools*** they have or their use.



Common cognitive impairments in neuro populations

- Attention
- Memory
- Speed of processing
- Executive function (high level skills)
- Apraxia

What do you think the physiotherapist's role in the assessment and rehab of cognition?

What problems may affect your physiotherapy session?



MDT working and cognition

Psychology

- Psychometric assessments
- Evaluate mood

Physiotherapy
Physical restoration



Doctors
Diagnostics

Occupational Therapy
Assess impact on occupational performance

Nursing
Care

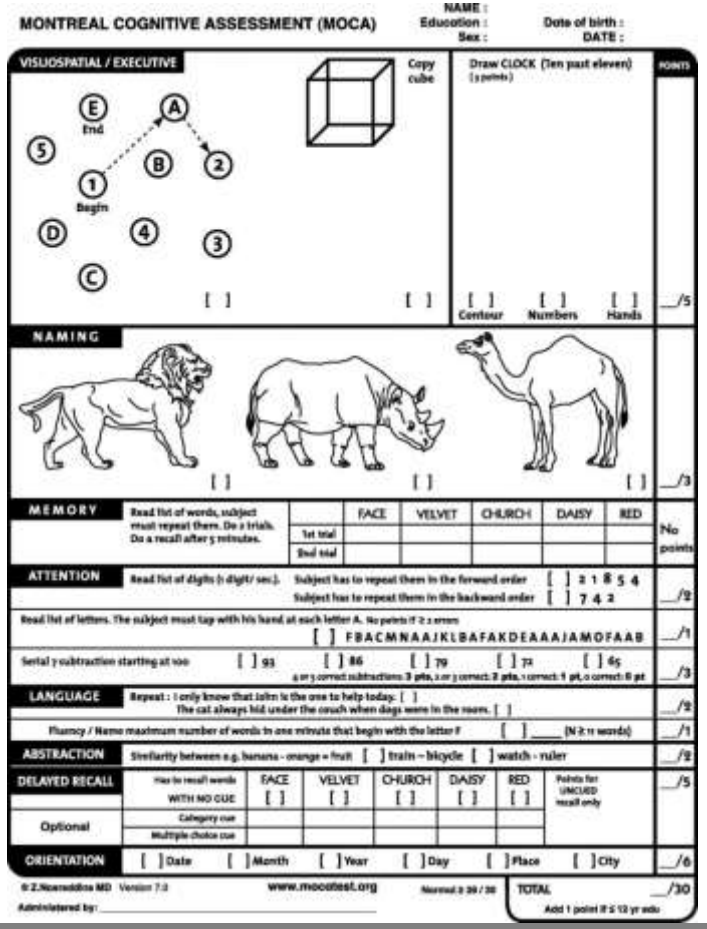
How do you assess cognitive and visual perceptual impairment?

- Cognitive and perceptual screens
- Functional assessments
- In-depth standardised assessment

Cognition and visual perceptual Screens

Montreal Cognitive Assessment (MoCA)

- Visual spatial
- Attention
- Language
- Executive dysfunction
- Memory



MONTREAL COGNITIVE ASSESSMENT (MOCA)

NAME :
Education :
Sex :
Date of birth :
DATE :

VISUOSPATIAL / EXECUTIVE — Copy cube — Draw CLOCK (ten past eleven) (3 points) — Contour — Numbers — Hands — /5

NAMING — /3

MEMORY — Read list of words, subject must repeat them. Do 2 trials. Do a recall after 5 minutes. — FACE — VELVET — CHURCH — DAISY — RED — 1st trial — 2nd trial — No points

ATTENTION — Read list of digits (1 digit/ sec.). Subject has to repeat them in the forward order [ ] 2 1 8 5 4 — Subject has to repeat them in the backward order [ ] 7 4 2 — /2

Read list of letters. The subject must tap with his hand at each letter A. No points if ≥ 2 errors [ ] F B A C M N A A J K L B A F A K D E A A A J A M O F A A B — /1

Serial 7 subtraction starting at 100 [ ] 93 [ ] 86 [ ] 79 [ ] 72 [ ] 65 — 4 or 5 correct subtractions: 3 pts, 2 or 3 correct: 2 pts, 1 correct: 1 pt, 0 correct: 0 pt — /3

LANGUAGE — Repeat : I only know that John is the one to help today. [ ] — The cat always hid under the couch when dogs were in the room. [ ] — /2

Fluency / Name maximum number of words in one minute that begin with the letter F [ ] ____ (N ≥ 11 words) — /1

ABSTRACTION — Similarity between e.g. banana - orange = fruit [ ] train – bicycle [ ] watch - ruler — /2

DELAYED RECALL — Has to recall words — FACE — VELVET — CHURCH — DAISY — RED — Points for UNCUED recall only — /5

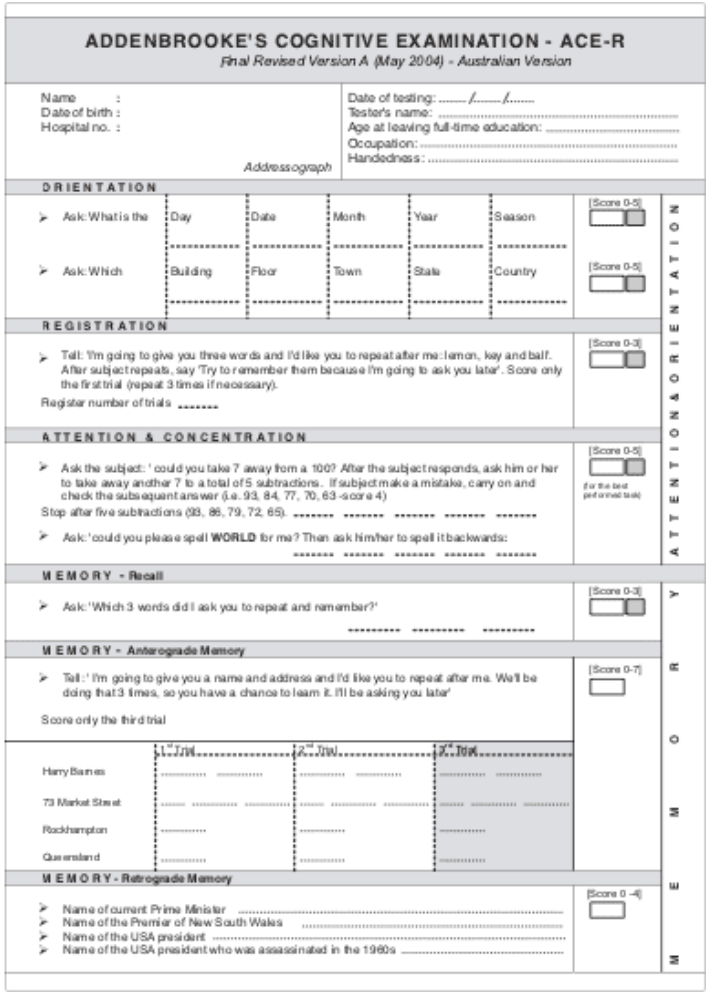
Optional — WITH NO CUE [ ] [ ] [ ] [ ] [ ] — Category cue — Multiple choice cue

ORIENTATION — [ ] Date [ ] Month [ ] Year [ ] Day [ ] Place [ ] City — /6

© Z. Nasreddine MD Version 7.3 — www.mocatest.org — Normal ≥ 26 / 30 — Administered by: _____

TOTAL — /30 — Add 1 point if ≤ 12 yr edu

The ACE cognitive assessment refers to the Addenbrooke's Cognitive Examination (ACE-III), a 15-minute test used to screen for cognitive impairment, particularly dementia. It evaluates five cognitive domains: attention, memory, verbal fluency, language, and visuospatial abilities, with a higher score (out of 100) indicating better cognitive function



ADDENBROOKE'S COGNITIVE EXAMINATION - ACE-R

Final Revised Version A (May 2004) - Australian Version

Name :
Date of birth :
Hospital no. :

Addressograph

Date of testing: ...../...../.....
Tester's name: .....
Age at leaving full-time education: .....
Occupation: .....
Handedness: .....

ORIENTATION [Score 0-5]

- Ask: What is the: Day | Date | Month | Year | Season
- Ask: Which: Building | Floor | Town | State | Country [Score 0-5]

REGISTRATION [Score 0-3]

- Tell: 'I'm going to give you three words and I'd like you to repeat after me: lemon, key and ball. After subject repeats, say 'Try to remember them because I'm going to ask you later'. Score only the first trial (repeat 3 times if necessary).

Register number of trials .....

ATTENTION & CONCENTRATION [Score 0-5] (for the best performed task)

- Ask the subject: ' could you take 7 away from a 100? After the subject responds, ask him or her to take away another 7 to a total of 5 subtractions. If subject make a mistake, carry on and check the subsequent answer (i.e. 93, 84, 77, 70, 63 -score 4)

Stop after five subtractions (93, 86, 79, 72, 65). ....

- Ask: 'could you please spell **WORLD** for me? Then ask him/her to spell it backwards:

....

MEMORY - Recall [Score 0-3]

- Ask: 'Which 3 words did I ask you to repeat and remember?'

....

MEMORY - Anterograde Memory [Score 0-7]

- Tell: ' I'm going to give you a name and address and I'd like you to repeat after me. We'll be doing that 3 times, so you have a chance to learn it. I'll be asking you later'

Score only the third trial

	1st Trial	2nd Trial	3rd Trial
Harry Barnes	....	....	....
73 Market Street	....	....	....
Rockhampton	.....	.....	.....
Queensland	.....	.....	.....

MEMORY - Retrograde Memory [Score 0 -4]

- Name of current Prime Minister .....
- Name of the Premier of New South Wales .....
- Name of the USA president .....
- Name of the USA president who was assassinated in the 1960s .....

MEMORY **ATTENTION&ORIENTATION**

Cognitive observations

Can they focus on your instruction?

Can they concentrate over a long period or does their ability to listen/ follow what you say get more difficult over time

Do they forgot to attend appointment or forget advice you have given

Do they have difficulty copying your instructions or using objects

Do they take a long time to follow your instruction.

Visual perception observations

- Can they find objects at bed space?
- What can patients see within the room?
- Do they bump into items?
- Recognise their bedside and find their way around the ward?
- Can they describe how their body is organised?
- Can they recognise an objects?
- Difficulty judging distances- ie transferring from bed to chair?

Case study- Mr B (PART B)

- Reluctance to turn to the left hand side when walking
- Overfilled a cup of water at the bedside.
- Unable to find his way in familiar places when it is busy
- Find its difficult to hold a conversation when it is busy.
- Doesn’t always listen to instructions

What kind of cognitive and perceptual abilities do you think has be affected by the stroke?



Cognitive and visual perceptual rehabilitation

4 principles:

1. **Education**; ‘drip feed’ approach; lots of repetition.

2. **Process/impairment based;** targeting specific impairments with exercises to overcome the problem.

3. **Strategy/ compensatory;** training to compensate for the problem.

4. **Function;** practice the skills learned in their everyday life.

Memory impairment- interventions in physiotherapy

- Write down important information
- Reinforcement and repetition
- External aids – prompt sheets, pictures
- Consistency – location, tasks, methods

•Preparation – revisit before session, remind at end

•Consider location for therapy

•Create links to aid memory – e.g. diary

•Ask family and MDT colleagues for Strategies

•Be aware of how this limits rehabilitation

Attention deficits- interventions in physiotherapy

- Give instruction one step at a time
- Reduce distractions
- Grade complexity of tasks
- Simple instructions
- Identify limits of tolerance/fatigue
- Timing of sessions

Speed of Processing: interventions in physiotherapy

- Identify delay – count
- Identify patterns – worse with fatigue
- Allow time for information to be received/response- WAIT!
- Simple instructions

Disorientation- Interventions in physiotherapy:

- Re-orientate patient
- State clear aims
- Repetition of information – consider format
- Use environmental cues
- Consider environment for rehab, e.g. home vs hospital
- Familiar items, people and places may help – use family
- Consider MDT intervention, e.g. orientation boards

Apraxia - Interventions in physiotherapy:

- Consider context of object use
- Task-specific training with strategies
- Structure and familiarity
- Hand-over-hand facilitation
- Prevention of errors

Dysexecutive impairment- Interventions in physiotherapy:

- External strategies – lists, prompt sheets, alarms, pictures
- Give feedback
- Prevent errors in performance

Insight- Intervention in Physiotherapy

- Education
- Reinforcement/repetition
- Evaluation of performance, pre- and post sessions

Visual and Spatial Perception

Assessment: observation, formal testing

Interventions in Physiotherapy:

- Identify problem – vision? depth?
- Consider contrast in environment
- Minimise obstacles

•Practice and repetition
•Liaise with MDT colleagues
•Compensatory techniques

Neglect/Inattention- Intervention in Physiotherapy:

Observation; Do they bump into items, turn their head to affected side, can they find objects on affected side

- Follow general MDT recommendations, eg. approach from affected side, increase stimuli on affected side
- Reinforce scanning
- Activities crossing midline
- Shifting attention between left and right

Topographical disorientation- Intervention in physiotherapy

<u>Assessment</u>

- Observation
- Repeat route finding? Any new learning



- Follow general MDT recommendations
- Picking out and remembering key landmarks in the environment
- Use signs – entries and exit signs.
- Encouraging patient to vocalise directions

Agnosia- Intervention in Physiotherapy:

Observation; can the person recognise objects visually? By touch?

- Education
- Labelling
- Hand patient objects required
- Liaise with MDT colleagues for advice
- Hand over hand facilitation
- Use same objects, e.g. same ball, same frame

Key messages

- Cognitive and visual perceptual impairments are common in most neurological conditions and will require physiotherapist to adapt your approach to rehab
- Importance of observation and assessment
- Importance of MDT working
- Importance of understanding strategies to assist with rehab, note these may change if progressive in nature

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